

Conditional Statements

▶ A **CONDITIONAL STATEMENT** is a logical statement using the words “IF” and “THEN”

▶ Example:

IF I do my chores, **THEN** I get my allowance.



Conditional Statements

▶ There are two parts to Conditional Statements:

- ▶ The **HYPOTHESIS** (the **IF** part)
- ▶ The **CONCLUSION** (the **THEN** part)

▶ Example:

▶ IF I do my chores, THEN I get my allowance.



Converse, Inverse, Contrapositive

▶ Recall our original Conditional Statement

If I **do my chores**, then I **get my allowance**

- ▶ Using this Conditional, we can write three other statements
 - ▶ Converse
 - ▶ Inverse
 - ▶ Contrapositive



Converse

▶ The **CONVERSE** is formed by switching the hypothesis and conclusion (**SWITCH**)

Original Conditional $p \longrightarrow q$
If I **do my chores**, then I **get my allowance**

Converse $q \longrightarrow p$
If I **get my allowance**, then I **did my chores**

▶ Notes Examples



BICONDITIONALS

▶ When a conditional statement and its converse are both true, the two statements can be combined.

▶ Use the phrase **IF AND ONLY IF** (abbreviated: **IFF**)

▶ Symbolic Notation

▶ $p \longleftrightarrow q$

▶ Remember, $p \longrightarrow q$ AND $q \longrightarrow p$ BOTH must be true!



BICONDITIONAL

▶ Example

▶ **Conditional:** If an angle is right, then it has a measure of 90°.
▶ True!

▶ **Converse:** If an angle has a measure of 90°, then it is right.
▶ True!

▶ **Biconditional:** An angle is right iff it measures 90°.
▶ An angle measures 90° iff it is right.



Inverse

▶ The **INVERSE** is formed by negating the hypothesis and the conclusion of the original statement (**NEGATE**)

Original Conditional $p \longrightarrow q$
If I **do my chores**, then I **get my allowance**

Inverse $\sim p \longrightarrow \sim q$
If I **my DON'T do my chores**, then I **DON'T get my allowance**

▶ Notes Examples



Contrapositive

▶ The **CONTRAPOSITIVE** is formed when you negate the converse (**SWITCH AND NEGATE**)

Original Conditional $p \longrightarrow q$
If I **do my chores**, then I **get my allowance**

Contrapositive $\sim q \longrightarrow \sim p$
If I **DON'T get my allowance**, then I **DIDN'T do my chores**

▶ Notes Examples



Negation

▶ A statement can be altered by negation

- ▶ Doing the **OPPOSITE**

▶ The symbol for negation is **~**

▶ Example

- ▶ Statement: We are in school
- ▶ Negation: We are NOT in school

▶ Notes Examples



BICONDITIONALS

▶ NON-EXAMPLE

▶ **Conditional:** If we are in Geometry class, then we are in school.
▶ True!

▶ **Converse:** If we are in school, then we are in Geometry class.
▶ Not always true!

▶ **Can't be written as a BICONDITIONAL!**



Symbolic Notation

▶ Conditional Statements can be written in Symbolic Notation

- ▶ The **HYPOTHESIS** is marked by the letter **p**
- ▶ The **CONCLUSION** is marked by the letter **q**

▶ Example

- ▶ p: “I do my chores”
- ▶ q: “I get my allowance”



Translating English to Mathematics

▶ English:

- ▶ IF I do my chores, THEN I get my allowance

▶ Mathematics:

- ▶ Let p be “I do my chores”
- ▶ Let q be “I get my allowance”
- ▶ $p \Rightarrow q$
- ▶ Read “p implies q”



Summing It Up

- ▶ Converse
 - ▶ SWITCH!
- ▶ Inverse
 - ▶ NEGATE!
- ▶ Contrapositive
 - ▶ SWITCH AND NEGATE!

